

Data Intake Processing and Verification Report

Background Information

- **Original Dataset Name:** Global Mangrove Distribution, Aboveground Biomass, and Canopy Height
- **GHG Center Dataset Title:** Global Mangrove Aboveground Biomass and Canopy Height
- **Dataset Provider:** NASA
- **Date Obtained:** May 2025
- **Location Obtained from:** <https://doi.org/10.3334/ORNLDAAAC/1665>
- **Data Location in GHG Center:** cms-mangrove-agb-canopyheight-grid-v1.3
- **Data POC(s):** Dr. Lola Fatoyinbo, Dr. Marc Simard
- **Dataset File Type(s):** GeoTIFF
- **Projection (if different from WGS84):** NA

Data Transfer Confirmation

An MD5 checksum is used to detect high-level errors within data transmissions

- Results for individual checksum file comparisons of pre-transfer and post-transfer for a few files are shown below:

Filename	MD5 Original file	MD5 Downloaded file
Mangrove_agb_AndamanAndNicobar.tif	b5dc0d78deb20a985ffbed0ce0cbe353	b5dc0d78deb20a985ffbed0ce0cbe353
Mangrove_hmax95_AndamanAndNicobar.tif	3e8b9f6897ac2650d99bd12e58365e7b	3e8b9f6897ac2650d99bd12e58365e7b
Mangrove_hba95_AndamanAndNicobar.tif	a0a17ad3b69d471cde020bc78c5e084e	a0a17ad3b69d471cde020bc78c5e084e

- All files were transferred successfully
- Report any individual file issues: N/A

Data Intake Process

- https://us-ghg-center.github.io/ghgc-docs/data_workflow/cms-mangrove-agb-canopyheight-grid-v1.3_Data_Flow.html

Dataset Statistics (subset):

GeoTIFF (original data) and COGs (transformed Cloud Optimized GeoTIFF) files for 2 randomly chosen dates were read to generate statistics to validate transformed data against the original data files.

- Statistics for 2 random files for each variable are summarized below:

Above ground biomass (agb) [Unit: Mg ha⁻¹]

Filename (Original/COG)	Minimum	Maximum	Mean	Standard Deviation
Mangrove_agb_AndamanAndNicobar.tif	0.4575	827.5093	304.6976	229.5626
Mangrove_agb_AndamanAndNicobar_2000year.tif	0.4575	827.5093	304.6976	229.5626
Mangrove_agb_Australia.tif	57.4102	212.6169	120.1786	38.2071
Mangrove_agb_Australia_2000year.tif	57.4102	212.6169	120.1786	38.2071

Height (hmax95) [Unit: meters]

Filename (Original/COG)	Minimum	Maximum	Mean	Standard Deviation
Mangrove_hmax95_Gabon.tif	0.8485	62.7890	23.1636	18.0174
Mangrove_hmax95_Gabon_2000year.tif	0.8485	62.7890	23.1636	18.0174
Mangrove_hmax95_NorthernMarianalIslands.tif	10.1820	50.9099	42.3735	8.7281
Mangrove_hmax95_NorthernMarianalIslands_2000year.tif	10.1820	50.9099	42.3735	8.7281

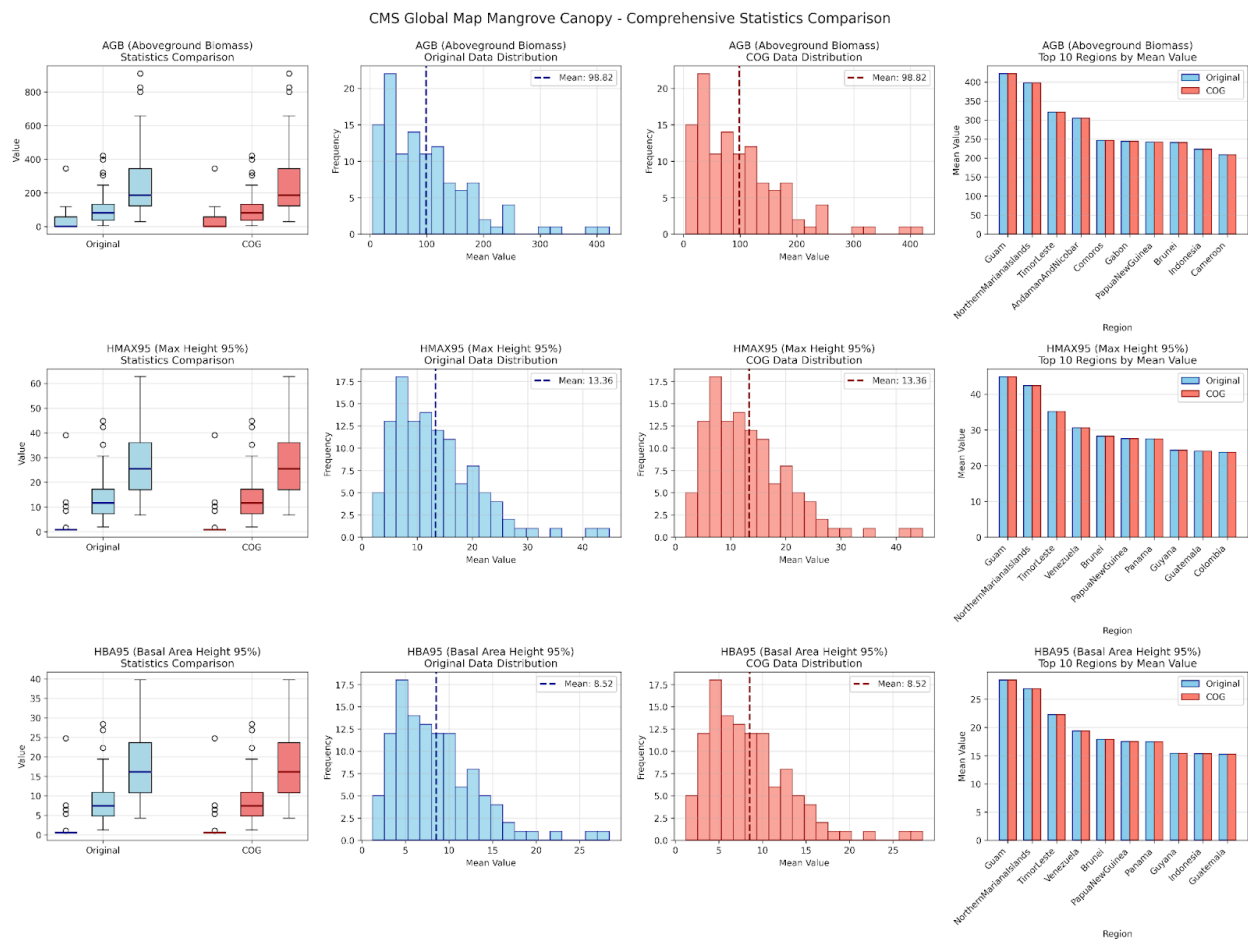
Basal area weighted height (hba95) [meters as units]

Filename (Original/COG)	Minimum	Maximum	Mean	Standard Deviation
Mangrove_hba95_CaribbeanCaymanIslands.tif	0.5377	11.8294	7.0645	2.3233

Mangrove_hba95_CaribbeanCaymanIslands_2000year.tif	0.5377	11.8294	7.0645	2.3233
Mangrove_hba95_NorthernMarianalIslands.tif	6.4524	32.2620	26.8524	5.5311
Mangrove_hba95_NorthernMarianalIslands_2000year.tif	6.4524	32.2620	26.8524	5.5311

Overall Dataset Statistics Visualization

- More detailed statistics for a few files are shown below (randomly chosen)



Summary

- We are confident that the transformation and display of data in the US GHG Center accurately represents the source data

- Link to the transformation code
- Link to the dataset Tutorial
- Link to the US GHG Center data catalog page

Report Completed on: 7/15/2025

MSFC POC for questions: [Jeanné le Roux](#), [Siddharth Chaudhary](#)